

Q1.

- (a) Matthew is finding a formula for the inverse function $\operatorname{arsinh} x$.
He writes his steps as follows:

$$\text{Let } y = \sinh x$$

$$y = \frac{1}{2}(e^x - e^{-x})$$

$$2y = e^x - e^{-x}$$

$$0 = e^x - 2y - e^{-x}$$

$$0 = (e^x)^2 - 2ye^x - 1$$

$$0 = (e^x - y)^2 - y^2 - 1$$

$$y^2 + 1 = (e^x - y)^2$$

$$\pm \sqrt{y^2 + 1} = e^x - y$$

$$y \pm \sqrt{y^2 + 1} = e^x$$

To find the inverse function, swap x and y : $x \pm \sqrt{x^2 + 1} = e^y$

$$\ln(x \pm \sqrt{x^2 + 1}) = y$$

$$\operatorname{arsinh} x = \ln(x \pm \sqrt{x^2 + 1})$$

Identify, and explain, the error in Matthew's proof.

- (b) Solve $\ln(x + \sqrt{x^2 + 1}) = 3$

(2)

(1)

(Total 3 marks)

Q2.

- (a) Use the definitions of $\sinh x$ and $\cosh x$ in terms of e^x and e^{-x} to show that

$$x = \frac{1}{2} \ln \left(\frac{1+t}{1-t} \right) \text{ where } t = \tanh x$$

(4)

- (b) (i) Prove $\cosh^3 x = \frac{1}{4} \cosh 3x + \frac{3}{4} \cosh x$

(4)

- (ii) Show that the equation $\cosh 3x = 13 \cosh x$ has only one solution.

Find this solution in exact logarithmic form.

(4)
(Total 12 marks)

Q3.

(a) Prove that $\frac{\sinh \theta}{1 + \cosh \theta} + \frac{1 + \cosh \theta}{\sinh \theta} \equiv 2 \coth \theta$

Explicitly state any hyperbolic identities that you use within your proof.

(4)

(b) Solve $\frac{\sinh \theta}{1 + \cosh \theta} + \frac{1 + \cosh \theta}{\sinh \theta} = 4$ giving your answer in an exact form.

(2)

(Total 6 marks)

Q4.

(a) Using the definition $\tanh y = \frac{e^y - e^{-y}}{e^y + e^{-y}}$, show that, for $|x| < 1$,

$$\tanh^{-1} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

(3)

(b) Hence, or otherwise, show that $\frac{d}{dx} (\tanh^{-1} x) = \frac{1}{1-x^2}$.

(3)

(c) Use integration by parts to show that

$$\int_0^{\frac{1}{2}} 4 \tanh^{-1} x \, dx = \ln \left(\frac{3^m}{2^n} \right)$$

©2025 Exam Papers Practice. All rights reserved.

where m and n are positive integers.

(5)

(Total 11 marks)

Q5.

(a) Sketch the graph of $y = \tanh x$.

(2)

(b) Given that $u = \tanh x$, use the definitions of $\sinh x$ and $\cosh x$ in terms of e^x and e^{-x} to show that

$$x = \frac{1}{2} \ln \left(\frac{1+u}{1-u} \right)$$

(6)

(c) (i) Show that the equation

can be written as

$$3 \tanh^2 x - 7 \tanh x + 2 = 0$$

(2)

(ii) Show that the equation

$$3 \tanh^2 x - 7 \tanh x + 2 = 0$$

has only one solution for x .

Find this solution in the form $\frac{1}{2} \ln a$, where a is an integer.

(5)

(Total 15 marks)

Q6.

(a) Express

$$5 \sinh x + \cosh x$$

in the form $Ae^x + Be^{-x}$, where A and B are integers.

(2)

(b) Solve the equation

$$5 \sinh x + \cosh x + 5 = 0$$

giving your answer in the form $\ln a$, where a is a rational number.

(4)

(Total 6 marks)

©2025 Exam Papers Practice. All rights reserved.

Q7.

(a) Given that

$$4 \cosh^2 x = 7 \sinh x + 1$$

find the two possible values of $\sinh x$.

(4)

(b) Hence obtain the two possible values of x , giving your answers in the form $\ln p$.

(3)

(Total 7 marks)

Q8.

The curve C has equation

$$y = \cosh x - 3 \sinh x$$

- (a) (i) The line $y = -1$ meets C at the point $(k, -1)$.

Show that

$$e^{2k} - e^k - 2 = 0$$

(3)

- (ii) Hence find k , giving your answer in the form $\ln a$.

(4)

- (b) (i) Find the x -coordinate of the point where the curve C intersects the x -axis, giving your answer in the form $p \ln a$.

(4)

- (ii) Show that C has no stationary points.

(3)

- (iii) Show that there is exactly one point on C for which $\frac{d^2 y}{dx^2} = 0$.

(1)

(Total 15 marks)